

AirInsight: Data-Driven AQI, Health, and Policy Exploration Tool

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1 INTRODUCTION

This project will create an interactive air quality visualization platform that allows users to explore current and historical Air Quality Index (AQI) by United States (U.S.) county. Users may overlay AQI data with other environmental and health factors, such as electric vehicle (EV) charging stations and air pollution-related health incidents. As an advanced feature, we will build a model to empower users to simulate how policy interventions like adding EV stations could impact regional air quality and predict the air quality level in the future.

2 PROBLEM DEFINITION

Air pollution is a major, preventable cause of illness and premature death. Not only does air pollution lead to lung-related illnesses such as COPD and lung cancer, but it also contributes to other preventable diseases affecting the heart, eyes, bones, and brain [15]. Air pollution is a worldwide issue, attributed to causes such as vehicle emissions [11], manufacturing [9], and agriculture [6]. Furthermore, exposure to Particle Matter (PM) 2.5 and ozone has been linked to adverse birth outcomes, including preterm birth and low birth weight [3]. A recent study in Australia [1] concluded that air pollution remains a material, preventable cause of death, with over half of PM2.5 and NO2 mortality burden being tied to traffic-related pollution. The goal of these studies was to identify the issue, rather than find a solution.

Existing research has focused on identifying the problem rather than providing actionable, spatially targeted solutions. An interactive tool that enables policymakers or the public to visualize localized air quality alongside related health and environmental factors does not exist today, nor the ability to model how interventions could improve future air quality outcomes.

3 LITERATURE SURVEY

Contemporary studies often compare air quality levels before and after an intervention. However, such changes are difficult to measure due to noisy data. One study [2] highlights that air quality and health outcomes require careful consideration of weather and seasonal influences. While relevant to our modeling

approach, the study overlooks infrastructure changes. Another study assessed the "Clean Air Zone" policy and bus fleet upgrades introduced in Leeds, United Kingdom [14]. The study successfully identifies changes in PM concentrations, but lacks data pre-processing, limiting its ability to isolate effects. A further study uses geolocation and regression to visualize PM2.5 concentration in China [10]. This supports our use of spatial mapping to track pollution, but lacks integration of external drivers such as infrastructure, which our project aims to address through layered, dynamic visualizations.

Lastly, a study by the Polytechnic University of Catalonia (UPC) in Barcelona [16] showed significant potential for increased usage of EVs to improve air quality in Barcelona and Madrid, but their methodology used HERMESv2.0 [8] which the authors describe as "a high-resolution emission model specifically developed for Spain". Since our project seeks a flexible approach enabling adhoc analysis of the potential impact of anti-pollution measures on U.S. regions, the use of ecoregions, as described in [12], to generalize results of anti-pollution measures across regions was explored.

Accurate air quality measurement is essential when analyzing air quality trends, particularly in regions where monitoring coverage is sparse. Prior studies have demonstrated that averaging air quality values across ecologically or demographically similar regions can substantially reduce bias compared with relying on a single fixed-site monitor. Both unweighted and population-weighted spatial averaging have been found to produce stronger correlations with true PM2.5 exposure fields and reduce measurement error [17] [7]. Further, the use of spatial aggregation based on ecoregions was supported as a scientifically valid approach to generating representative exposure estimates across broader geographic units [19]. In addition to spatial methods, temporal interpolation techniques have been shown to preserve temporal trends in pollutant concentrations while addressing missing data limitations [4] [13]. These studies collectively justify the use of ecoregion-based averaging and linear interpolation as robust and widely accepted imputation techniques in air quality research, forming the basis for the methodological framework adopted in this project.

4 PROPOSED METHOD

4.1 Data Collection and Integration

Data to support this project was downloaded from eight data sources and stored in a Google Cloud Platform (GCP) bucket. The data sources, size, and number of records are shown in Table 1.

Data Source	Size	Records
Daily AQI readings by county	294 MB	125,280
EV charging stations	188 MB	537,086
Health indicators	280 MB	99,904
U.S. Population	89 MB	358,417
U.S. EPA Ecoregions	42 MB	3,114
Custom U.S. county map file	842 KB	N/A
County-state unique identifiers	18 MB	3,144
Daily temperature by county	1.11 GB	3,616,978

Table 1: Data Sources

Figure 1 visualizes the distribution of AQI data completeness by county.

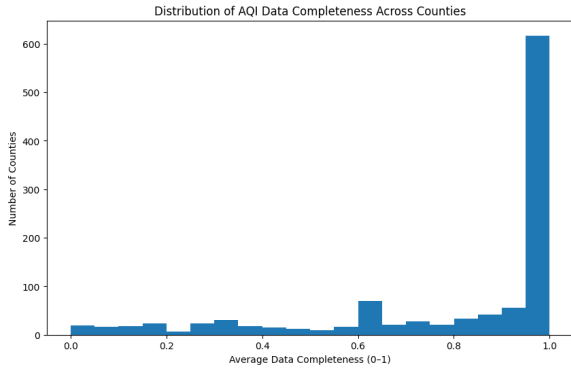


Figure 1: Distribution of AQI Data Completeness

To ensure reliability and minimize exposure misclassifications, counties with less than 30% data completeness or more than 180 consecutive days of missing data were excluded from the analysis. Additionally, county month pairs with less than 50% data availability were flagged for imputation. Missing daily AQI values were imputed using the mean of other counties within the same EPA ecoregion, as such approach was shown to be effective [17] [18] [7] [19]. This leverages ecological homogeneity to produce regionally representative estimates. When insufficient daily data were available within an ecoregion, gaps were filled using bilinear

interpolation, a method supported by [4] and [13] as effective in preserving temporal variability while addressing missing records.

Following imputation, daily values were aggregated to monthly averages for each county, resulting in a final dataset of 928 counties suitable for analysis. Figure 2 shows the proportion of counties with available AQI data within each state. States with few counties, such as the District of Columbia and Delaware, exhibit almost complete coverage. In contrast, most states have data for fewer than 50% of their counties, with Nebraska showing the lowest coverage at 9%.

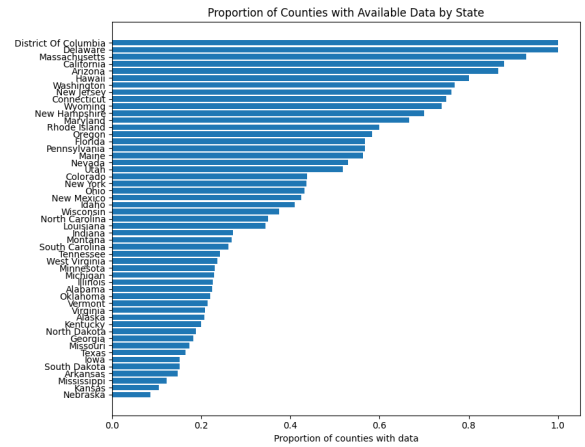


Figure 2: Proportion of Counties with AQI Data

Figure 3 displays the trends in AQI for the five counties with the highest and lowest average AQI values. The most polluted counties include San Bernardino, Riverside, Los Angeles, Kern (all California), and Maricopa (Arizona), while the least polluted counties are Aleutians East (Alaska), Apache (Arizona), Routt (Colorado), Wilson, and Karnes (Texas). The counties with the highest AQI exhibit clear seasonal patterns, with AQI levels increasing during the spring and peaking in the summer months.

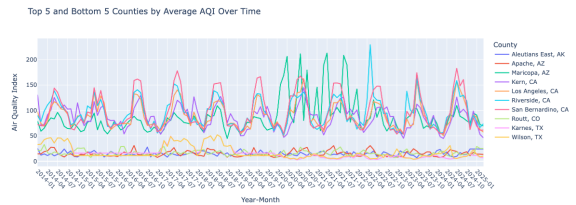


Figure 3: Top and Bottom AQI Counties

4.2 Intuition

Building on the concept introduced by Gagliardi [5], **our project is innovative in three ways**: First, by capturing the pollutant of greatest concern on a given day at the county level. Second, by applying the Holt-Winters exponential smoothing method, our model explicitly accounts for seasonality and long-term trends, which many baseline models either ignore or approximate poorly. Third, by incorporating EV charging infrastructure as a novel predictor, to explore how adoption contributes to air quality outcomes. Unlike traditional predictors, EV stations are a modern intervention, and linking them to AQI trends offers a fresh perspective on environmental change. By combining established modeling techniques with unconventional data, our project delivers novel and actionable insights into the dynamics of air pollution.

4.3 Holt-Winters Exponential Smoothing

An additive Holt-Winters exponential smoothing model was implemented to capture the level, trend, and seasonality time-series components in the monthly AQI data. The additive formulation assumes that the amplitude of seasonal fluctuations remains approximately constant over time. Using the Statsmodels Python library, the model optimizes smoothing parameters automatically to minimize forecast error.

The model was trained on all but the most recent 12 months of historical data per county, and its predictive accuracy evaluated on the most recent 12 months for out-of-sample forecasting. The method also computes a 95-percent confidence interval based on forecast residual variance. Lastly, the accuracy is quantified using RMSE and MAE, and the time-series components are exported for interpretability. The trained models are used to project AQI 12 months into the future.

4.4 EV Modeling

Our EV analysis consists of three approaches: (1) an exploratory study of EV charging infrastructure impacts, (2) five Bayesian regression models, and (3) a two-way fixed-effects regression model.

For each county, we assemble a dataset containing 2024 mean AQI, population density, EV charging station density, county land area, and geographic attributes.

This step allows us to assess EV station density effects on AQI and to identify confounders, such as the tendency for EV adoption to vary systematically with population density or regional characteristics.

In addition, we implemented two-way fixed-effects regression to control for county fixed effects (α_i), which capture unchanging characteristics such as elevation and landforms, and year fixed effects (T_t) that affect every county in a given year (e.g. COVID, economic cycles). The model can be represented by:

$$AQI_{it} = \beta_1 EV_{it} + \beta_2 Temp_{it} + \alpha_i + \tau_t + \varepsilon_{it}$$

where:

- EV_{it} is EV charging stations per 100,000 residents
- $Temp_{it}$ is the average October temperature
- α_i represents county fixed effects
- τ_t represents year fixed effects

Using this model, we generated predicted AQI values under hypothetical increases in EV infrastructure. This gives a practical, policy-focused view of what increasing EV adoption could mean for local air quality.

4.5 Data Visualization

The data visualizations are developed in Power BI, integrated directly with Google BigQuery data views. The dashboard features three tabs, each with interactive maps for (1) Pollution-related health indicators, (2) AQI forecasts (Figure 4), and (3) simulated EV infrastructure effects on AQI (Figure 6). When a user hovers over a county, the tooltip displays both the actual and predicted AQI for that location (Figure 5). All views allow the user to interact with the data and/or the algorithms through clicking and hovering actions.

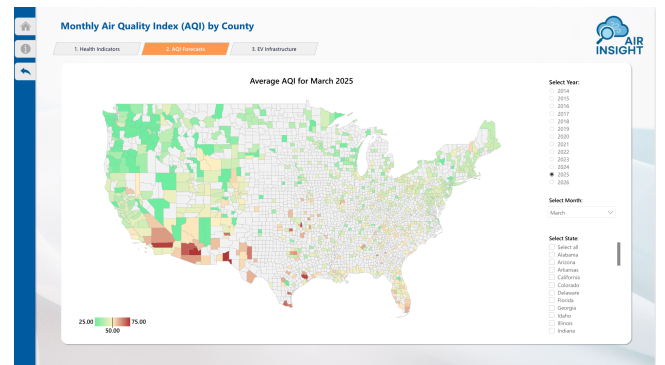


Figure 4: Historic and Predicted AQI

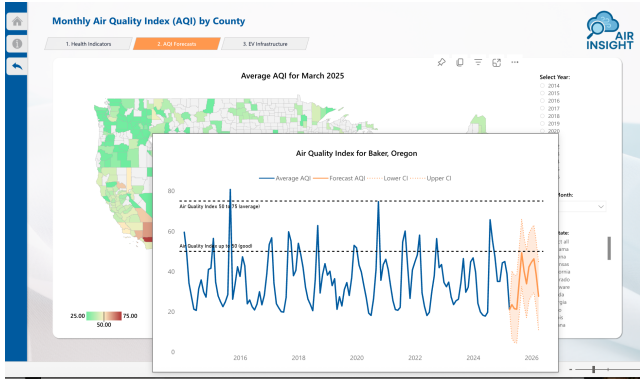


Figure 5: Dynamic Tooltips

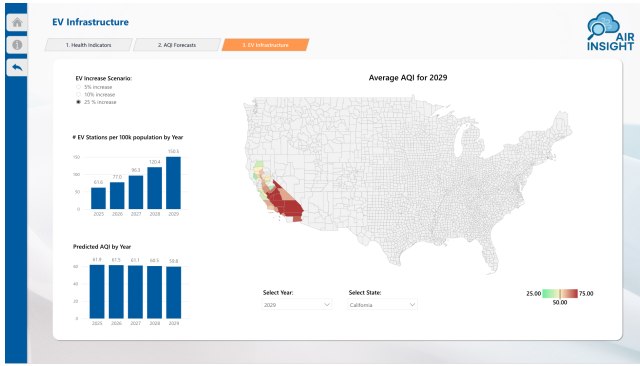


Figure 6: EV Station Impact on Future AQI

5 EVALUATION

5.1 Holt-Winters Evaluation

After fitting the additive Holt-Winters model on the cleaned county-level AQI data, the visualization displays a 12-month forecast with a 95-percent confidence interval for each county from April 2025 to March 2026. Figure 7 shows the HW forecast broken down into temporal components for Baldwin County, AL.

Baldwin County's air quality showcases clear annual seasonality and moderate month-to-month variability. The HW decomposition indicates that AQI fluctuations are driven primarily by seasonal effects rather than structural shifts. The baseline level declines steadily until around 2021, after which it begins to rise again in 2022, suggesting a mild rebound in underlying AQI levels. Similarly, the trend component shows a gradual upward trajectory beginning in 2022, indicating a significant shift toward slightly worsening air quality in recent years that may warrant continued monitoring.

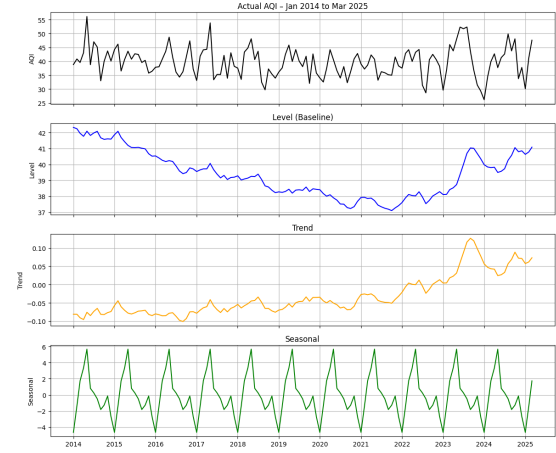


Figure 7: HW Component Plots for Baldwin, AL

To evaluate robustness, a time-series train-test split was applied for every county, where the 12 most recent months were reserved as the test set. This one-year horizon assesses the model's short-term predictive reliability on unseen data.

Across the hundreds of counties analyzed, the average forecast performance resulted in an RMSE of around 6.9 AQI units and a MAE of 5.5 AQI units, shown in Table 2. The relationship between RMSE and MAE is useful in understanding the severity of outliers, and their relative proximity suggests that the errors are reasonably consistent. Given that each AQI category (as defined by the EPA) spans 50 points (e.g. 51-100 = Moderate), the model's forecasts typically remain within the same air-quality category as the actual observations. In rare cases, such as during unexpected wildfires or extreme meteorological anomalies, predictions may deviate more substantially, but such events are inherently difficult to anticipate using historical patterns alone.

Statistic	RMSE	MAE
count	928	928
mean	6.92	5.48
stdev	4.37	3.36
min	1.98	1.66
25%	4.43	3.57
50%	5.92	4.69
75%	8.14	6.40
max	57.97	51.18

Table 2: Holt-Winters Forecast Error Summary

Overall, the results suggest that the Holt-Winters model provides stable, interpretable forecasts with a level of precision suitable for practical air-quality visualization and planning purposes.

5.2 EV Modeling Evaluation

The analysis focuses on 2024, the year with complete EV station data, and combined AQI, demographic, political, and geographic datasets.

5.2.1 Cross-sectional view of AQI vs. EV charging stations. The dataset is constructed where, for each county:

- 2024 mean AQI
- population density (using an Albers equal-area conic projection)
- EV charging station density
- Percentage of Democratic votes in the 2024 presidential election
- county area and geospatial boundaries

AQI was normalized by log-population density, and plotting this AQI against EV station density results in a moderate negative correlation of $R = -0.38$, which indicates that more EV charging stations may lower AQI (cleaner air). However, an additional step is taken to address possible limitations such as confounding factors. EV station density is also found to be correlated to Democratic-voter density, where counties with a higher percentage of Democratic voters have more EV stations ($R = 0.35$ and 0.68 , without and with log transformation of EV station density). In addition, AQI was lower for counties with a higher percentage of Democratic voters. This illustrates the belief that counties with stronger environmental policy preferences both adopt EV infrastructure more rapidly and implement other pollution-reducing policies.

5.2.2 Bayesian Regression. Another approach we tried was using Bayesian regression models. We fit 5 models with the response as $\log(\text{AQI})$ and temperature and EV density predictors:

- (1) Simple lognormal regression
- (2) Hierarchical model
- (3) Hierarchical model adding year impact
- (4) Hierarchical model adding year and ecoregion impact
- (5) Hierarchical model adding year, ecoregion, and state impact

Our results showed the fourth model performed best with a negative coefficient for EV impact, however it is not statistically significant. Therefore, our focus shifts towards the FE model for clearer interpretations.

5.2.3 Panel Data Fixed Effects Regression. To better isolate the relationship between EV charging infrastructure and air quality, we estimate a two-way fixed-effects model on the 2014–2024 panel. County fixed effects account for all characteristics that do not change over time, while yearly fixed effects remove nationwide influences like wildfire seasons or unusually warm years.

Across the different versions of the model, the coefficient on EV station density remains small but consistently negative.

In the first and most basic model includes only EV density and fixed effects, adding one station per 100,000 residents is linked to a 0.049-point decrease in AQI ($p = 0.0012$). Adding temperature into the model reduces the estimated effect slightly (to -0.0375), and the relationship remains statistically significant. Including lagged AQI or lagged EV infrastructure does not seem to change the result significantly: the lagged terms are not significant, and the EV effect stays negative.

	FE	FE+Temp	Final
EV per 100k	-0.049*** (0.015)	-0.038** (0.013)	-0.021** (0.011)
Temp	–	0.392*** (0.068)	0.302*** (0.057)
Entity FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes
Obs.	4595	4595	4595
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$			

Table 3: Two-Way FE Regression Results

We also ran the model separately on different clusters of counties by running separate models for population size, population density, climate zone, and baseline pollution levels. The clearest improvements in AQI show up in counties with mid-level population density and moderate climates. Conversely, rural areas, densely populated urban areas, or counties with extreme climate conditions show insignificant change. This suggests that the influence of EV infrastructure may depend on local environmental and the people living there.

To complement the regression results, we use the estimated coefficients and fixed effects to project AQI under three EV-growth scenarios. We simulate annual EV station increases of 5%, 10%, and 25% for the years 2025–2029 and compute predicted AQI for each county. The resulting trajectories generally show modest reductions in AQI as EV infrastructure expands, though the magnitude varies across counties depending on local environmental and demographic conditions. As an illustration, Figure 8 displays projected AQI paths for Mono County, California, under the three growth scenarios.

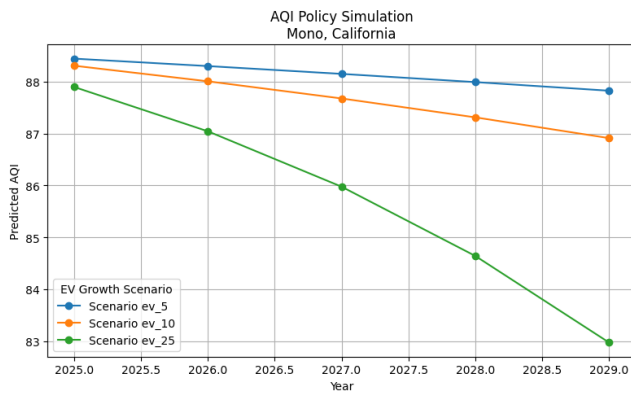


Figure 8: Projected AQI in EV Growth Scenarios

5.2.4 Limitations. The fixed-effects model cannot capture county factors that change over time, and only adjusts for national shocks. The only meteorological factor included is temperature, and EV growth is slow in earlier years, making the estimates weaker.

5.3 Data Visualization Evaluation

Our final product delivers a user interface that allows a user to interact with eight cleaned and two algorithm-based (AQI forecasts and AQI predictions based on EV station increases) datasets. Users can use the application to view current and forecasted future AQI based on U.S. county, and they can also simulate how policy interventions such as additional EV stations may improve air quality for a given state or county

6 CONCLUSIONS & DISCUSSION

The project objective was to create an interactive air quality visualization to enable policymakers and the

public to make informed, data-driven decisions and explore interventions that could improve future air quality outcomes.

One challenge we have overcome during this project is the varying time-ranges, level of detail, and scope of the different datasets. For example, AQI data is consistently available across 70%+ counties by month over many decades, with daily changes. In contrast, EV data is static until a new EV charging station is built and operationalized, at which point the data changes. We have addressed this issue by modeling the dynamic AQI data on top of the static EV data.

If research were to continue on this project, we would look to address the following items:

- (1) *Additional EV Data:* EV charging stations remain quite sparse, therefore we would like to expand our EV data to include EV ownership by county in addition to existing infrastructure.
- (2) *Political Views:* While we began to explore the relationship between EV stations and political ideology, we would like to dig deeper, specifically looking at the strength of air-quality regulation policies that underly the relationship of AQI with political ideology.
- (3) *Expanded Meteorological Variables:* Temperature is important for explaining AQI patterns, but is not the only important meteorological variable (wind, humidity, etc.). By adding more weather factors, we gain deeper insights into how much EV stations matter relative to further weather changes.

Even facing challenges and potential for expansion, our current dashboard and predictive algorithms provide significant impact to the general public. The ability to not only visualize past AQI but also view predicted future AQI allows people to take air quality into account when determining where to live. It also allows both the general public and policymakers the ability to simulate how air quality may change if policy decisions, such as increased EV infrastructure, were to go into effect.

All team members have contributed a similar amount of effort throughout all aspects of the project.

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